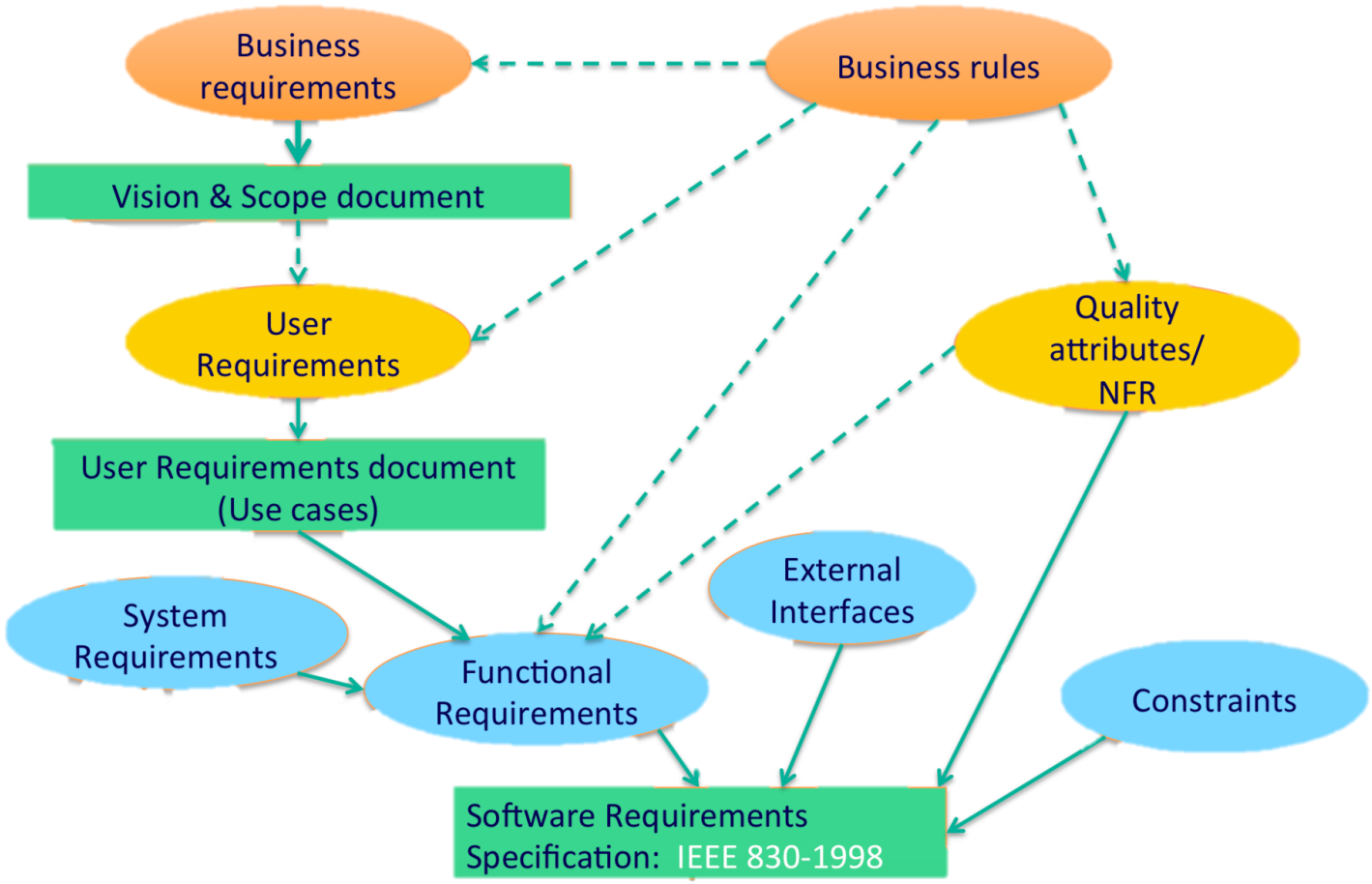


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SE2203
Software Requirements Engineering

Lecture 6:
Quality attributes. NFR.

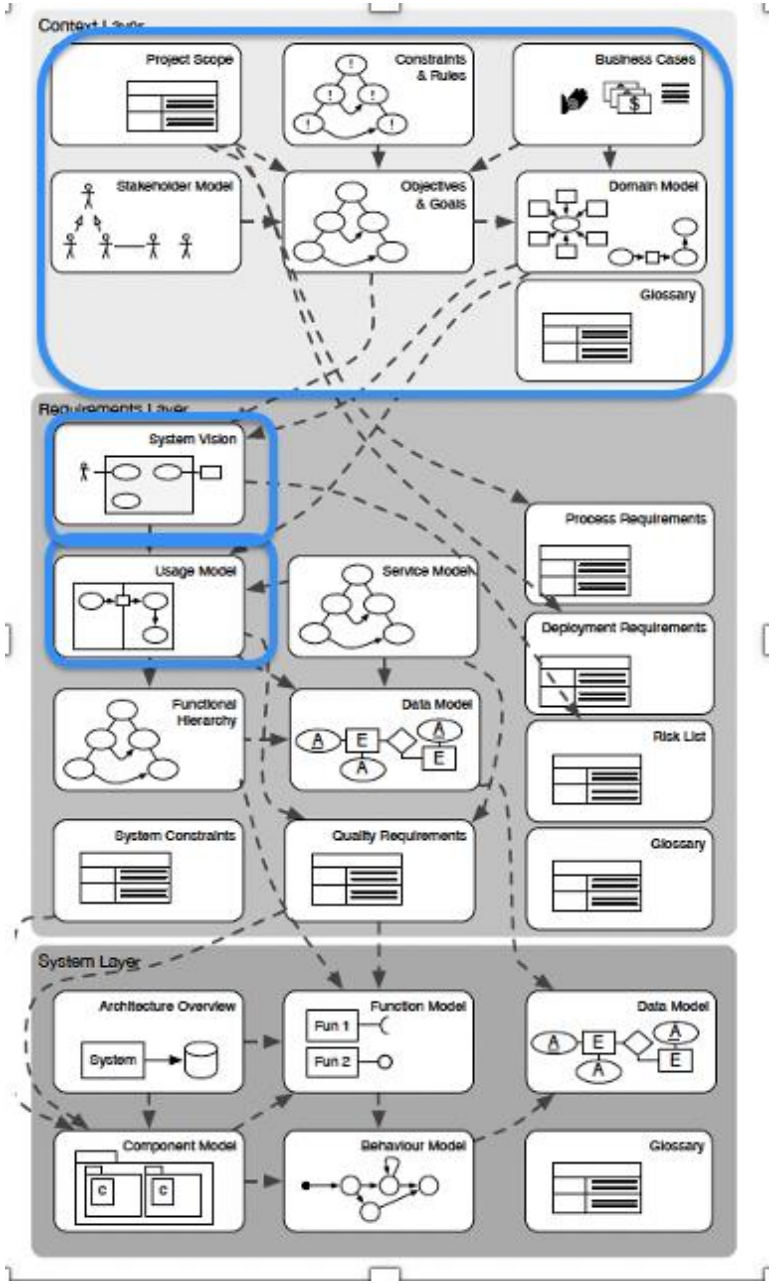
Recap time


Business requirements
Business rules
Vision & Scope document
User Requirements
Quality attributes/ NFR
User Requirements document (Use cases)
External Interfaces
System Requirements
Functional Requirements
Constraints
Software Requirements Specification: IEEE 830-1998

Recap time: AMDiRE Approach

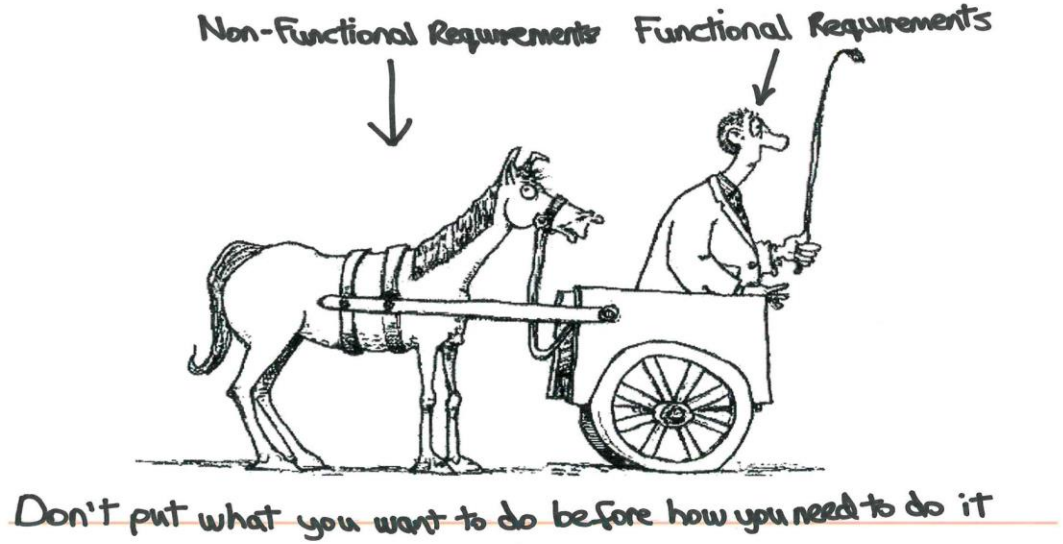
- D. Fernández, B. Penzenstadler (2015) Artefact--based Requirements Engineering: The AMDiRE Approach

Vision & Scope



Learning outcomes for Lecture 6

- What is product quality?
- What are quality attributes?
- What are non-functional requirements?



What is product quality?

- Quality might be determined subjectively
- Users don’t like to have their routine

destroyed by a “new and improved” system

Q: New Skype feedback

Split from this thread.

It's the most disgusting looking update ever - it's like a child's version and all my functionality is lost - I liked having a list of "Online" people - "Active Now" is similar but there's no way of telling their mood status like before - can I roll back ?

Reply I have the same question (5) • • •





Software quality attributes

- Many product characteristics can be called quality attributes
- … but most projects require carefully consider only few of them – this depends on project
- There are many quality standards/ schemes:
  - IEEE 1998
  - ISO 9000:2000
  - ISO/IEC 2007
  - ISO/IEC 2011
  - …
- Qualitative attributes often cannot be verified directly

- **ISO = International Organisation for Standardisation** (international standard-setting body composed of representatives from various national standards organisations, founded in 1947)
- **IEC = International Electrotechnical Commission** (International Standards Organisation that prepares and publishes International Standards for all electrical, electronic and related technologies, founded in 1906)
- **IEEE = Institute of Electrical and Electronics Engineers** (professional association founded in 1963)

External quality (1)

Primarily important to users
- "Quality in use"
- Related to behaviour and functional usage



Availability: The extent to which the system's services are available when and where they are needed



Performance: How quickly and predictably the system responds to user inputs or other events



Reliability: How long the system runs before experiencing a failure



Robustness: How well the system responds to unexpected operating conditions (errors), recovers from problem situations, forgiving users mistakes



Safety: How well the system is protected against doing any injury or damage



Security: How well the system protects against unauthorized access to the application and its data

External quality (2)

Primarily important to users
- "Quality in use"
- Related to behaviour and functional usage



Usability: How easy it is for people to learn, remember, and use the system.



Interoperability: How easily the system can interconnect and exchange data with other systems or components.



Installability: How easy it is to install, uninstall, and reinstall the application correctly.



Integrity: The extent to which the system protects against data inaccuracy and loss.

Might happen that some quality requirements logically fit to more than one category:
don't repeat them in all categories.

- Why this is important?

Internal quality

- Significant to development and maintenance staff
- Indirectly contribute to customer satisfaction by making the product easier to enhance, correct, test, and migrate to new platforms

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Reusability: To what extent components can be used in other systems



Maintainability: How easy it to maintain the system (**Modularity**)



Modifiability: How easy it is to [maintain,] change, enhance, and restructure the system



Scalability: How easily the system can grow to handle more users, transactions, servers, data, etc.



Verifiability: How easily developers and testers can confirm that the software is implemented correctly



Efficiency: How efficient the system uses computer resources



Portability: How easily the system can be made to work in other operating environments

Quality attributes for particular system types

- Certain attributes are of particular importance in certain system types
  - **Embedded systems**: reliability, robustness, safety, security, performance, efficiency, usability 4
  - **Web and corporate applications**: availability, scalability, performance, integrity, security, usability, interoperability 4
  - **Desktop and mobile systems**: usability, security, performance



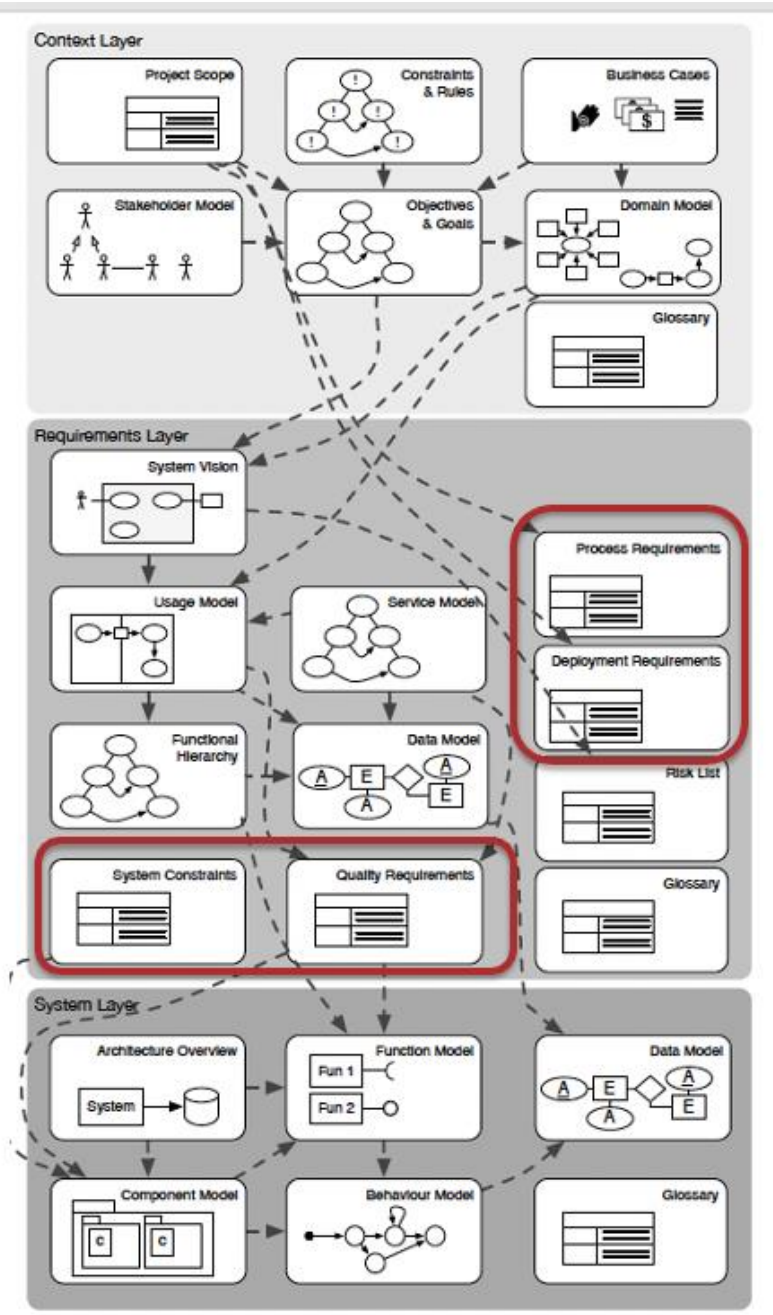
Functional vs. non-functional

- **Functional requirements** describe the functionality of the system (its functional behavior)
- **Non-functional requirements (NFR)** describe “everything that is non functional”:
  - Quality attributes
  - Constraints
    - Implementation constraints (operating system, programming languages, third-party components, …)
    - Deployment constraints
    - System constraints
  - Process requirements: characteristics of process/project (model of the development process, documentation, development standards, plan, costs, …)

AMDiRE Approach: NFR

D. Fernández, B. Penzenstadler (2015) Artefact--based Requirements Engineering: The AMDiRE Approach



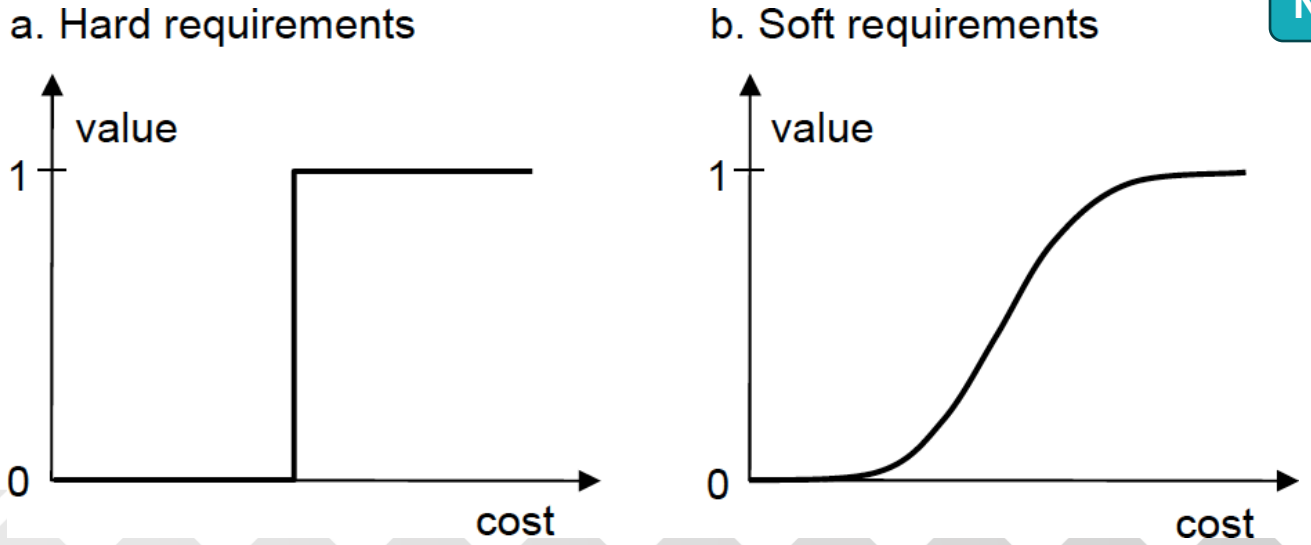


Hard vs. soft requirements

Similar to hard/soft goals

- **Hard requirement** is either completely satisfied or it is not satisfied
- Soft requirements can be gradually satisfied, i.e., the degree of satisfaction is measured on a scale which is at least ordinal

NFR are often soft



Refinement abstract requirements

- “The new call center application shall be easy to use.”
- How to measure and refine?

- **Functional solution**: The system shall have a help function that provides the user with support at any time and any functionality.

- **Non-functional solution/measure**: The call center staff member [the typical user] understands the core functionalities of the application after 10 minutes of training and can apply them at least as efficiently as the functionalities of the old application.

Quality attributes: Analysis

1. **List a rich set of quality attributes to consider**
2. **Reduce the list:**
   - Analyze the stakeholders' views
   - Record the rationale for deciding that a particular quality attribute is in or out for consideration
3. **Prioritize the attributes (e.g., using Brosseau's spreadsheet)**
   - Priorities of users might be very different from priorities of maintenance staff and technical support!
4. **Elicit specific expectations for each attribute:**
   - What the users are thinking when they say the software must be user-friendly, fast, reliable, robust, …

Quality attributes: Analysis

5. Specify well-structured quality requirements:

- **SMART**: **S**pecific, **M**easurable, **A**ttainable, **R**elevant, **T**ime-sensitive
- Simplistic requirements are not useful, as they provide little guidance and are not measurable,

  e.g.,

“The system shall be user-friendly – to subjective and vague

“The system shall be available 24/7” – rarely realistic or necessary

Brosseau's spreadsheet: Example

What is the highest possible score?

- Sample quality attribute prioritization for an airport check-in system

	Score	Availability	Integrity	Performance	Reliability	Robustness	Security	Usability	Verifiability
Availability	2		^	^	^	<	^	^	<
Integrity	6			<	<	<	^	<	<
Performance	4				<	<	^	^	<
Reliability	2					<	^	^	^
Robustness	1						^	^	<
Security	7							<	<
Usability	5								<
Verifiability	1								

"If I could have only one of these attributes, which would I take?"

Eliciting specific expectations

Roxanne E. Miller. (2009).
The Quest for Software Requirements

- Users won't know how to answer questions like "What are your performance requirements?", "How reliable should be the software?"
- There are guidelines/ list of questions that might help to elicit the corresponding information, e.g.,
  - What would be a reasonable or acceptable response time for retrieval of information in response to a query?
  - **What would users consider an unacceptable response time for a typical query?**
  - How many simultaneous users do you expect on an average?
  - What is the maximum number of simultaneous users that you would expect?
  - What times of the day, week, month, or year have much heavier usage than usual?

You can also analyze and decompose the quality goals to the level of measurable requirements

Requirement patterns

Active area of applied research:

Stephen Withall. (2007). software requirement patterns

- **Christopher Alexander. The Timeless Way of Building:**

“each pattern describes a problem which occurs over and over again in our environment, and then describes the core of the solution to that problem, in such a way that you can use this solution a million times over, without ever doing it the same way twice”

L. Barcelos, R, Penteado. (2017). Elaboration of software requirements documents by means of patterns instan9tiation

- **Approach of Barcelos & Penteado (2017):**

Template, which has a text with fixed and variable (parameter) parts Example: The "Execution Condition" pattern provides the template

- " The < operation > should only be allowed if < condition > “

NFR template

ID	Unique identifier
NFR	NFR description
Goal / Rationale	Goal, from which the requirement was derived
Satisfaction criteria	Metrics that have to be achieved to meet requirement, and the corresponding steps to measure/determine
Risk	Risks that might incur if the requirement is not met

Availability

- Measure: UpTime / (UpTime + DownTime)
- More formally: MTBF / (MTBF + MTTR)
  - Mean Time Between Failures (MTBF)
  - Mean Time To Repair (MTTR) the system after failure
- Scheduled maintenance periods also affect availability
- Availability is
  - Strongly affected by the maintainability
  - Related to reliability

- Beware of specifying 100% as the expected value, as it will be impossible to achieve.
- Live-critical applications (e.g., air traffic control) might have “five 9’s requirement”: the system must be available 99.999% of the time, i.e., the system could be down no more than 5 minutes and 15 seconds per year.

Availability: What to consider?



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- How many hours in a given period the system must be available?
- Are there any periods for which availability is imperative to meet business or safety objectives?
- What modules/functionalities/… of the system are most critical for being available?
- What are the business consequences of the system being unavailable for users?
- If scheduled maintenance must be performed periodically, when should it be scheduled?
- What are the minimum and maximum durations of the maintenance periods?
- If maintenance activities must be performed while the system is up, what impact will they have on availability and how the impact can be minimized?
- What user notifications are necessary if the system becomes unavailable?

Availability requirements: Examples

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- **AVL-1:** The system shall be at least 95% available on weekdays between 6:00am and midnight Eastern Time, and at least 99% available on weekdays between 3:00pm and 5:00pm Eastern Time.

- **AVL-2**: Downtime that is excluded from the calculation of availability consists of maintenance scheduled during the hours from 6:00pm Sunday Eastern Time through 3:00am Monday Eastern Time.
- What might be an issue with the above requirements?

Performance

- Poor performance is annoying if the user waiting for a query results
- Might also represent serious risks to safety
- Depends on external factors: speed of the computer being used, network connections, other hardware components, …

Performance dimension	Example
Response time	Number of seconds to display a webpage
Throughput	Credit card transactions processed per second
Data capacity	Maximum number of records stored in a database
Dynamic capacity	Maximum number of concurrent users of a website
Predictability in real-time systems	Hard timing requirements for a flight-control system
Behavior in degraded modes or overloaded conditions	A natural disaster leads to a massive number of emergency phone system calls

Performance requirements: Examples

- **PER-1:** Authorization of an ATM withdrawal request shall take no more than 2 seconds.
- **PER-2:** Webpages shall be fully downloaded in an average of 3 seconds or less over a 30 megabits/second Internet connection
- **PER-3:** The system shall update the transaction status display within 1 second after the completion of each booking.



Reliability

- Measures:
  - Percentage of operations that completed correctly
  - Mean Time Between Failures (MTBF)
  - Maximum acceptable probability of a failure during a given time period
- High reliability requires high verifiability to make it easier to find defects that could compromise reliability
- Badly handled exceptions can make an impression of poor reliability

Reliability: What to consider?



- How would you judge whether the system was reliable enough?
- What would be consequences of experiencing a failure when performing certain operations within the system?
- What would you consider as a critical failure?
- What would you consider as a minor failure?
- If the system goes down, how long could it stay offline before it significantly affects your business operations?
- [also relates to availability]

Reliability requirements: Examples

- **REL-1:** No more than 5 experimental runs out of 1,000 can be lost because of software failures

- **REL-2:** The mean time between failures of the card reader component shall be at least 90 days.



Robustness

- Software errors have to be handled in a way the user perceives as reasonable, not annoying
  - Fault tolerance (e.g., are the user input errors caught and corrected?)
  - Survivability (e.g., can the hardware experience a drop from a certain height without damage?)
  - Recoverability (e.g., can the device resume proper operation if it loses power in the middle of the operation?)

- Analyze and ask users about error conditions the system might encounter, and how the system should react
- Analyze how to detect possible faults that could lead to the system failure, report them to the user, and recover from them if the failure occurs

Robustness requirements: Examples

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- **ROB-1**: If the text editor fails before the user saves the file, it shall recover the contents of the file being edited as of, at most, 1 minute prior the failure, next time the same user launches the application.

- **ROB-2**: All plot description parameters shall have default values specified, which the system shall use if a parameter's input data is missing or invalid.

Safety



- **Safety requirements**
  - Might be dictated by governmental regulations or business rules
  - Often are written in a form of condition or action the system must not allow to occur
  - Related to the hazard analysis
- Under what conditions might happen that a human is harmed by use of this product? How the system can detect these conditions? How should it respond?
- What is the maximum allowed frequency of failures that have the potential to cause injury?
- What failure modes have the potential of causing harm or damage?
- What operator actions have the potential of causing harm or damage?
- Are there specific modes of operation that pose risk to humans or property?

Safety requirements: Examples

- **SAF-1:** The user shall be able to see a list of all ingredients in any menu items, with ingredients highlighted that are known to cause allergic reactions.
- **SAF-2:** The system shall terminate any operation within 1 second if the measured boiler pressure exceeds 90% of the specified maximum pressure.
- **SAF-3:** The therapeutic radiation machine shall allow irradiation only if the proper filter is in place.



Therac--25: a radiation therapy machine involved in at least 6 accidents between 1985 and 1987.

Patients were given massive overdoses of radiation, resulting in death or serious injury.

And there are many other examples, unfortunately…

Security: What to consider?

- User authorization and privileges levels, user access control
- User identification/authentication: password construction rules, password change frequency, security questions, forgotten login/ password procedures, biometric identifications, account locking after unsuccessful access attempts, unrecognized computer, …
- Data privacy: who can create, see, change, copy, print, and delete what information
- Building audit trails of operations performed and access attempts
- What sensitive data has to be protected from an unauthorized access?
- What checks must be performed to confirm that the user is operation the application in a secure environment?
- Is there any specific user authentication method that must be used?

Security requirements: Examples

- **SEC-1:** The system shall lock a user's account after four consecutive unsuccessful login attempts within a period of 5 minutes.
- **SEC-2:** The system shall log all attempts to access secure data by users having insufficient privilege levels.
- **SEC-3:** The user shall have to change the temporary password assigned by the admin to a previously unused password immediately following the first successful login with the temporary password.



Usability

Active area of applied research

- Has many dimensions
- “Perceived ease of use” – habits, common functionalities, …

- What to analyze, includes:
  - The number of interactions (mouse clicks, touch-screen gestures, …) required to accomplish a task
  - The average time needed for a specific type of user to complete a particular task correctly
  - How many errors the user makes when completing the task
  - What percentage of a set of tasks the user can complete correctly without needing help

Usability requirements: Examples

- **USE-1:** A trained user shall be able to submit a request for a chemical from a vendor catalog in an average of 3 minutes, and in a maximum of 5 minutes.
- **USE-2:** All functions on the File menu shall have shortcut keys defined that the use the Control key pressed simultaneously with one other key.



Installability

- Installability covers the following activities:
  - Initial installation
  - Installation of a new version
  - Reverting to a previous version
  - Installation of additional components or updates
  - Recovery from an incomplete, incorrect, or user-aborted installation
  - Uninstallation
- Measure: the mean time to install the system
- Depends on a lot of factors:
  - How experiences the installer is
  - How fast the destination computer is
  - Medium from which the software is being installed

Examples of software installation:

- Downloading and updating apps to smartphone/tablet
- Installing and updating an OS
- Installing and updating a large commercial system
- Installing and updating an end-user application onto a desktop
- Downloading and updating software into a cable TV set-top box

Installability: What to consider?



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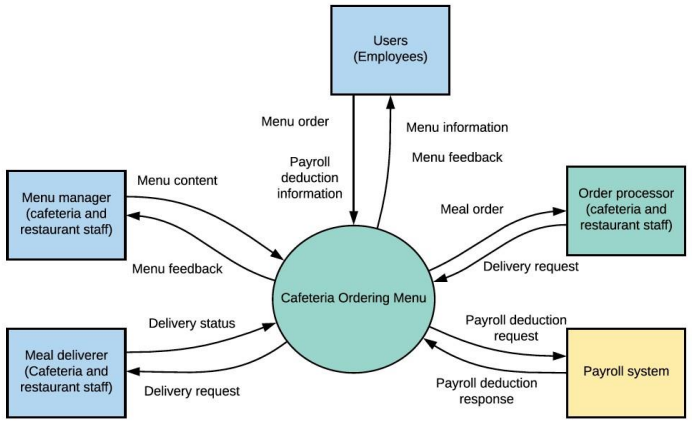
- What installation operations would require a restart of the application / computer/ device?
- What should the application do upon successful installation? What should the application do upon unsuccessful installation?
- What operations should be performed to confirm the validity of an installation?
- What other applications need to be shut down before performing the installation?
- What authorization or access privileges does the installer needed?
- How should the system handle an incomplete installation, such as one interrupted by a power failure or aborted by the user?

Installability requirements: Example

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- **INS-1:** An untrained user shall be able to successfully perform an initial installation of the application in an average of 10 minutes.
- **INS-2:** When installing an upgraded version of the application, all customizations in the user's profile shall be retained and converted to the new version's data format if needed.
- **INS-3:** The installation program shall verify the correctness of the download before beginning the installation process.
- **INS-4:** Installing this software on a server requires admin privileges.
- **INS-5:** Following successful installation, the installation program shall delete all temporary, backup and obsolete files associated with the application.

Interoperability



- Recall the context diagram
  - To what other systems should the SUD interface? What data/services must they exchange?
  - What standard data formats are necessary for the exchange?
  - What specific hardware components must interconnect with the system?
  - What messages or codes must the system receive and process from other system or devices?
  - What standard communication protocols are necessary to enable interoperability?
- Example:
- **IOP-1:** The AutoFocus-2 tool shall be able to import any valid AutoFocus-1 model.

Integrity: What to consider?

- Ensuring the persistence of changes that are made in the data
- Ensuring that changes in the data are made either entirely or not at all (backing out of a data change if a failure occurs through the operation)
- Coordinating changes made in multiple data stores, particularly when changes have to be made simultaneously
- Performing data backups:
  - At what frequency?
  - Automatically and/or on demand?
  - Of what data?
- Restoring data from a backup
- Archiving of data

Integrity requirements: Examples

- **INT-1:** After performing a file backup, the system shall verify the backup copy against the original and report any discrepancies.
- **INT-2:** The system shall protect against the unauthorized addition, deletion, or modification of data.



Questions?


HI, I'M ALAN, FROM THE QUALITY ASSURANCE DEPART-MENT.
DON'T BE WORRIED ABOUT QUALITY. I ASSURE YOU WE HAVE PLENTY OF IT.
THAT'S ALL YOU DO?
HEY, I DON'T TELL YOU HOW TO DO YOUR JOB.
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